Experiment 1

(1) Directory Operation

```
~/projects/linux2026 main*
❯ pwd
/home/jason/projects/linux2026

~/projects/linux2026 main*
❯ cd ~

~
❯ mkdir 0219

~
❯ cd 0219

~/0219
❯ mkdir cs

~/0219
❯ ls
cs

~/0219
❯ rmdir cs

~/0219
❯ ls
```

(2) File operation

```
~/0219
❯ cd ~

~
❯ touch 汪嘉诚.txt

~
❯ ls
0219                        Desktop                              miniforge3  PyCharmMiscProject
汪嘉诚.txt                  Documents                            Music       snap
audio_config_content.json   Downloads                            Pictures    Templates
common_config_content.json  games                                projects    video_config_content.json
config_info.json            linux-backup-2026-10-06_12-00-07.zip Public      Videos

~
❯ echo "hello world" > 0219.txt

~
❯ cat 0219.txt
hello world

~
❯ cp 汪嘉诚.txt 0219_copy.txt

~
❯ mv 汪嘉诚.txt 0219_newname.txt

~
❯ ls
0219                        config_info.json                     miniforge3          snap
0219_copy.txt               Desktop                              Music               Templates
0219_newname.txt            Documents                            Pictures            video_config_content.json
0219.txt                    Downloads                            projects            Videos
audio_config_content.json   games                                Public
common_config_content.json  linux-backup-2026-10-06_12-00-07.zip PyCharmMiscProject
```

(3)User Management

```
~
❯ whoami
jason

~
❯ sudo su root
[sudo] jason 的密码：
root@laptop-ubuntu:/home/jason# groupadd student
root@laptop-ubuntu:/home/jason# whoami
root
root@laptop-ubuntu:/home/jason# adduser NY0219
err: 请输入一个和配置变量 NAME_REGEX 的正则表达式相匹配的用户名。
使用"--allow-bad-names"选项来绕过这个检查或者重新配置
或重新配置 NAME_REGEX。
root@laptop-ubuntu:/home/jason# adduser --allow-bad-names NY0219
info: 允许使用可疑的用户名。
info: 正在添加用户"NY0219"...
info: 从 1000 到 59999 中选择 UID/GID...
info: 正在添加新组"NY0219" (1003)...
info: 正在添加新用户"NY0219" (1003) 到组"NY0219 (1003)"...
info: 创建主目录"/home/NY0219"...
info: 正在从"/etc/skel"复制文件...
新的密码：
无效的密码： 密码少于 8 个字符
重新输入新的密码：
抱歉，密码不匹配。
新的密码：
重新输入新的密码：
passwd: 已成功更新密码
正在改变 NY0219 的用户信息
请输入新值，或直接敲回车键以使用默认值
        全名 []:
        房间号码 []:
        工作电话 []:
        家庭电话 []:
        其它 []:
这些信息是否正确？ [Y/n] y
info: 正在添加新用户"NY0219"到附加组“users” ...
info: 正在添加用户"NY0219"到"users"组...
root@laptop-ubuntu:/home/jason# usermod -g student NY0219
root@laptop-ubuntu:/home/jason# su NY0219
NY0219@laptop-ubuntu:/home/jason$ cd ~
NY0219@laptop-ubuntu:~$ touch NY0219.txt
NY0219@laptop-ubuntu:~$ ls -l
总计 4
-rw-r--r-- 1 NY0219 student    0 10月  10 15:51 NY0219.txt
drwxr-xr-x 2 NY0219 student 4096 10月  10 15:49 模板
NY0219@laptop-ubuntu:~$
```

(4) Permission management

```
/tmp
❯ touch demo

/tmp
❯ ls -l demo
-rw-rw-r-- 1 jason jason 0 10月  10 15:58 demo

/tmp
❯ chmod 777 demo

/tmp
❯ ls -l demo
-rwxrwxrwx 1 jason jason 0 10月  10 15:58 demo

/tmp
❯ chmod a-x demo

/tmp
❯ ls -l demo
-rw-rw-rw- 1 jason jason 0 10月  10 15:58 demo

/tmp
❯ chmod go-w demo

/tmp
❯ ls -l demo
-rw-r--r-- 1 jason jason 0 10月  10 15:58 demo

/tmp
❯ chmod g+w demo

/tmp
❯ ls -l demo
-rw-rw-r-- 1 jason jason 0 10月  10 15:58 demo

/tmp
❯ chmod a=    demo

/tmp
❯ ls -l demo
---------- 1 jason jason 0 10月  10 15:58 demo

/tmp
❯ chmod +t demo

/tmp
❯ ls -l demo
---------T 1 jason jason 0 10月  10 15:58 demo

/tmp
❯ chmod u+s demo

/tmp
❯ ls -l demo
---S-----T 1 jason jason 0 10月  10 15:58 demo

/tmp
❯ chmod u+rx demo

/tmp
❯ ls -l demo
-r-s-----T 1 jason jason 0 10月  10 15:58 demo

/tmp
❯ chmod g+rws demo

/tmp
❯ ls -l demo
-r-srwS--T 1 jason jason 0 10月  10 15:58 demo
```

(5) Other management

```
~/demo
❯ mkdir others

~/demo
❯ cd others

~/demo/others
❯ touch a.txt b.txt

~/demo/others
❯ cd ..

~/demo
❯ tar -czvf backup.tar.gz others
others/
others/a.txt
others/b.txt

~/demo
❯ ls -l
总计 8
-rw-rw-r-- 1 jason jason  151 10月  10 16:08 backup.tar.gz
drwxrwxr-x 2 jason jason 4096 10月  10 16:08 others

~/demo
❯ tar -xzzvf backup.tar.gz
others/
others/a.txt
others/b.txt

~/demo
❯ ls -l
总计 8
-rw-rw-r-- 1 jason jason  151 10月  10 16:08 backup.tar.gz
drwxrwxr-x 2 jason jason 4096 10月  10 16:08 others
```

```
/tmp
❯ touch demo.txt

/tmp
❯ echo "no, I study in njupt.
this line ends with 00
this line ends with .00
I am 25 years old.
He is 15. It is 16.00
wy is the short of my name." > demo.txt


/tmp
❯ grep '^n' demo.txt
no, I study in njupt.

/tmp
❯ grep '\.00$' demo.txt

/tmp
❯ grep '\.00' demo.txt
this line ends with .00
He is 15. It is 16.00

/tmp
❯ grep '5\..' demo.txt
He is 15. It is 16.00

/tmp
❯ grep '^[wy]' demo.txt
wy is the short of my name.

/tmp
❯
```

assignments in lecture:

任务要求 1：

使用 openssh-server 连接虚拟机并完成基本命令操作

但我使用的是物理机，可以直接打开终端，安装 openssh-server 结果如下，可见之前就已经安装过了

```
~
❯ sudo apt install openssh-server
[sudo] jason 的密码：
正在读取软件包列表... 完成
正在分析软件包的依赖关系树... 完成
正在读取状态信息... 完成
openssh-server 已经是最新版 (1:9.6p1-3ubuntu13.19)。
```

后续如下:

```
jason@laptop-ubuntu /tmp/demo > pwd
/tmp/demo
jason@laptop-ubuntu /tmp/demo > ls
jason@laptop-ubuntu /tmp/demo > mkdir njupt
jason@laptop-ubuntu /tmp/demo > cd njupt
jason@laptop-ubuntu /tmp/demo/njupt > touch hello.txt
jason@laptop-ubuntu /tmp/demo/njupt > ls
hello.txt
jason@laptop-ubuntu /tmp/demo/njupt > echo "I love njupt" > hello.txt
jason@laptop-ubuntu /tmp/demo/njupt > cat hello.txt
I love njupt
jason@laptop-ubuntu /tmp/demo/njupt >
```

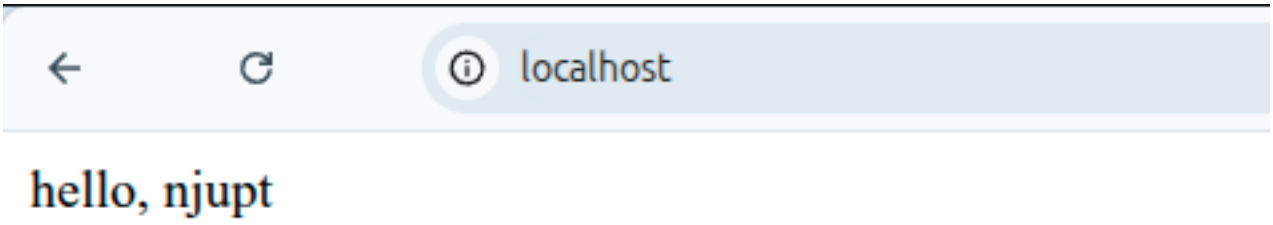
```
jason@laptop-ubuntu  /tmp/demo  pwd
/tmp/demo
jason@laptop-ubuntu  /tmp/demo  ls
jason@laptop-ubuntu  /tmp/demo  mkdir njupt
jason@laptop-ubuntu  /tmp/demo  cd njupt
jason@laptop-ubuntu  /tmp/demo/njupt  touch hello.txt
jason@laptop-ubuntu  /tmp/demo/njupt  ls
hello.txt
jason@laptop-ubuntu  /tmp/demo/njupt  echo "I love njupt" > hello.txt
jason@laptop-ubuntu  /tmp/demo/njupt  cat hello.txt
I love njupt
jason@laptop-ubuntu  /tmp/demo/njupt
```

放在/tmp 了因为不想污染目录

任务要求 2:

搭建 nginx 服务器，允许远程访问

```
$  sudo su root
root@laptop-ubuntu:/home/jason/projects/linux2026/B24060219 汪嘉诚 # cd /var/www/html/
root@laptop-ubuntu:/var/www/html# ls
index.nginx-debian.html
root@laptop-ubuntu:/var/www/html# vim index.html
```



因为是物理机所以是 localhost

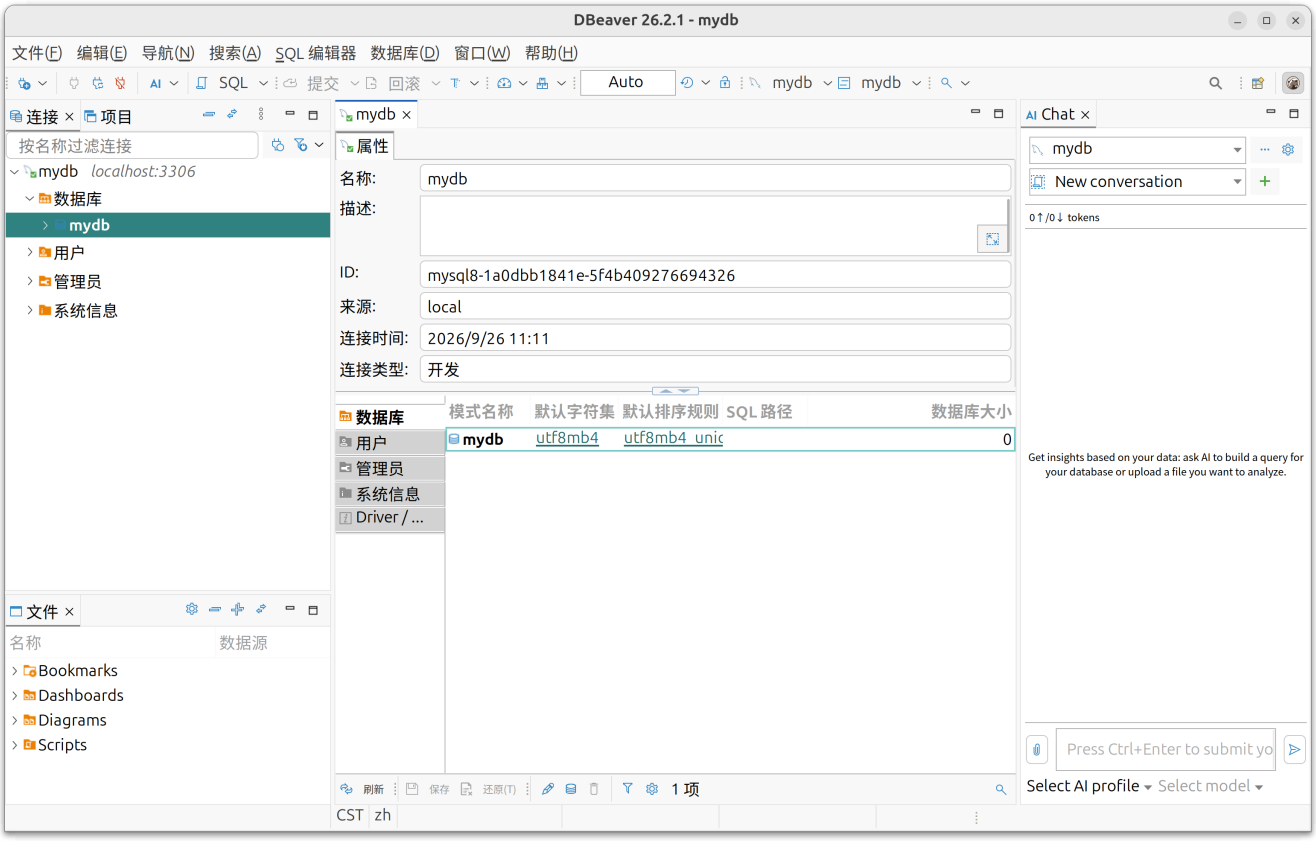
任务要求 3：

创建 mysql 服务器

```
mysql> CREATE DATABASE mydb CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
Query OK, 1 row affected (0.01 sec)
mysql> CREATE USER 'myuser'@'localhost' IDENTIFIED BY '111111';
Query OK, 0 rows affected (0.02 sec)
mysql> GRANT ALL PRIVILEGES ON mydb.* TO 'myuser'@'localhost';
Query OK, 0 rows affected (0.01 sec)
mysql> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.00 sec)
mysql> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mydb               |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
5 rows in set (0.00 sec)

mysql> exit
Bye
```

最后使用 dbeaver-ce 连接数据库，效果如图


DBeaver 26.2.1 - mydb
文件(F) 编辑(E) 导航(N) 搜索(A) SQL 编辑器 数据库(D) 窗口(W) 帮助(H)
连接 项目
按名称过滤连接
mydb localhost:3306
数据库
mydb
用户
管理员
系统信息
文件
名称
数据源
Bookmarks
Dashboards
Diagrams
Scripts
mydb
属性
名称: mydb
描述:
ID: mysql8-1a0dbb1841e-5f4b409276694326
来源: local
连接时间: 2026/9/26 11:11
连接类型: 开发
数据库
用户
管理员
系统信息
Driver / ...
模式名称 默认字符集 默认排序规则 SQL 路径 数据库大小
mydb utf8mb4 utf8mb4_unic 0
1 项
CST zh
Chat
mydb
New conversation
0↑/0↓ tokens
Get insights based on your data: ask AI to build a query for your database or upload a file you want to analyze.
Press Ctrl+Enter to submit yo
Select AI profile
Select model